



Problem

Given a string S, find the longest border of the string.

A border is a non-empty substring that is both a prefix and a suffix of the string, but it must be shorter than the entire string.

S = "ababa"

Prefixed: "a", "ab", "aba", "abab" Suffixes: "a", "ba", "aba", "baba"

The longest common prefix and suffix is "aba".

Therefore, the answer is:

aba

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Output Format

Print the longest border of S.

Sample Input 0

ababa

Sample Output 0

aba

[Change Theme](#)

Language

Python 3



```
1 # Enter your code here. Read input from STDIN.
2 s = input().strip()
3 n = len(s)
4 pi = [0] * n
5 j = 0
6 for i in range(1, n):
7     while j > 0 and s[i] != s[j]:
8         j = pi[j - 1]
9     if s[i] == s[j]:
10        j += 1
11    pi[i] = j
12 length = pi[-1]
13 print(s[:length])
14
```

Submissions

Leaderboard

Discussions

Line: 14 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Compiler Message

Test case 1

Success

